RATAN TATA MAHARASHTRA STATE SKILLS UNIVERSITY

School of Science Engineering and Technology Skills

Department of Engineering

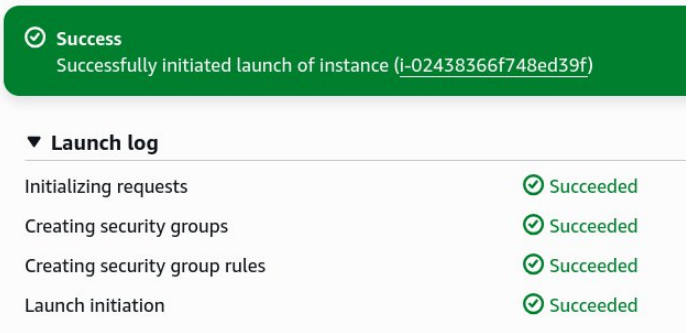
Subject – AWS Lab

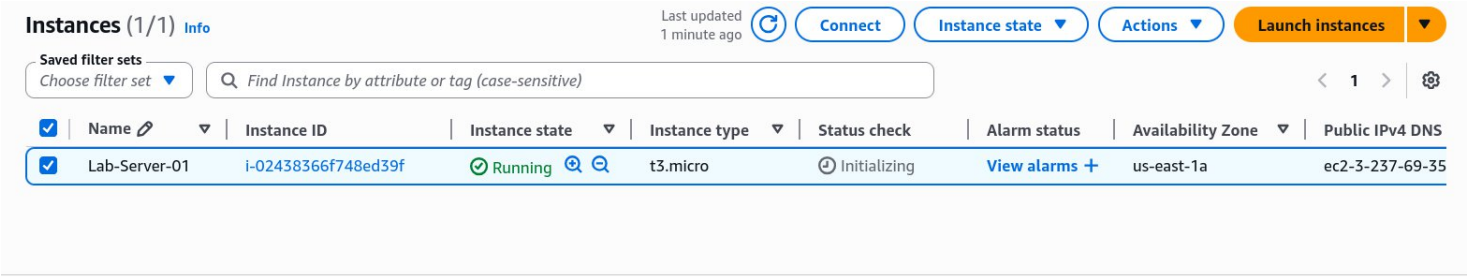
Experiment – 4

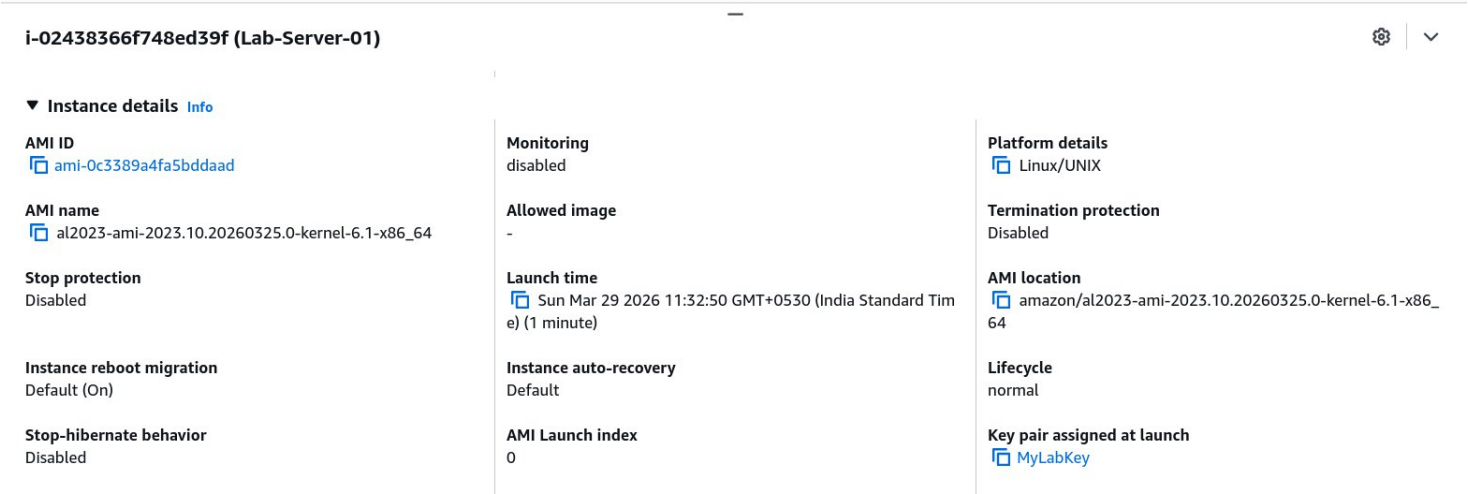
Lab Objective:
To provision a Linux virtual machine, establish a secure remote connection, perform administrative power operations, and manage persistent public identities using Elastic IPs.

Step 1: Launch an EC2 Instance

1. Navigate to the EC2 Console > Instances > Launch instances.
2. Name: Lab-Server-01
3. Application and OS Image (AMI): Select Amazon Linux 2023 AMI (Free Tier eligible).
4. Instance Type: Select t2.micro (or t3.micro depending on your region's Free Tier).
5. Key Pair: Click Create new key pair.
    a. Name: MyLabKey
    b. Format: .pem (for OpenSSH/Mac/Linux) or .ppk (for PuTTY/Windows).
    c. Download and save this file safely! You cannot download it again.
6. Network Settings: Select "Allow SSH traffic from My IP" (Security best practice).
7. Click Launch instance.







Step 2: Connect to Your Instance

There are two primary ways to access your Linux server:

Option A: EC2 Instance Connect (Browser-based)

1. Select your running instance in the console.
2. Click Connect > EC2 Instance Connect tab.
3. Click Connect. A terminal will open directly in your browser.

Option B: Standard SSH (Local Terminal)

1. Open your terminal and navigate to your key folder.
2. Set permissions (Linux/Mac only): chmod 400 MyLabKey.pem
3. Run the command:

   Bash:
   ssh -i "MyLabKey.pem" ec2-user@<Your-Public-IPv4-Address>

```
◄ 11s ❯ ssh -i MyLabKey.pem ec2-user@3.237.69.35
The authenticity of host '3.237.69.35 (3.237.69.35)' can't be established.
ED25519 key fingerprint is: SHA256:s6n6rVm13wSG9N3W3winMW4JqPOde09RnBWBGJxsBXw
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.237.69.35' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
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       _/m/'
[ec2-user@ip-172-31-14-165 ~]$ |
```
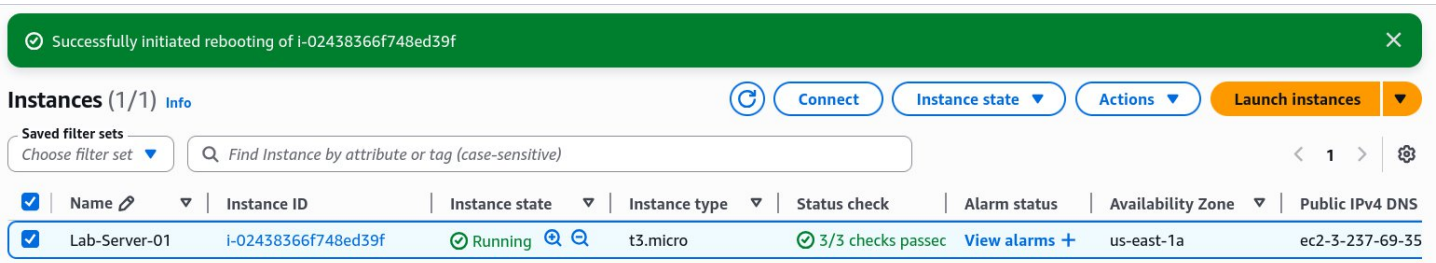
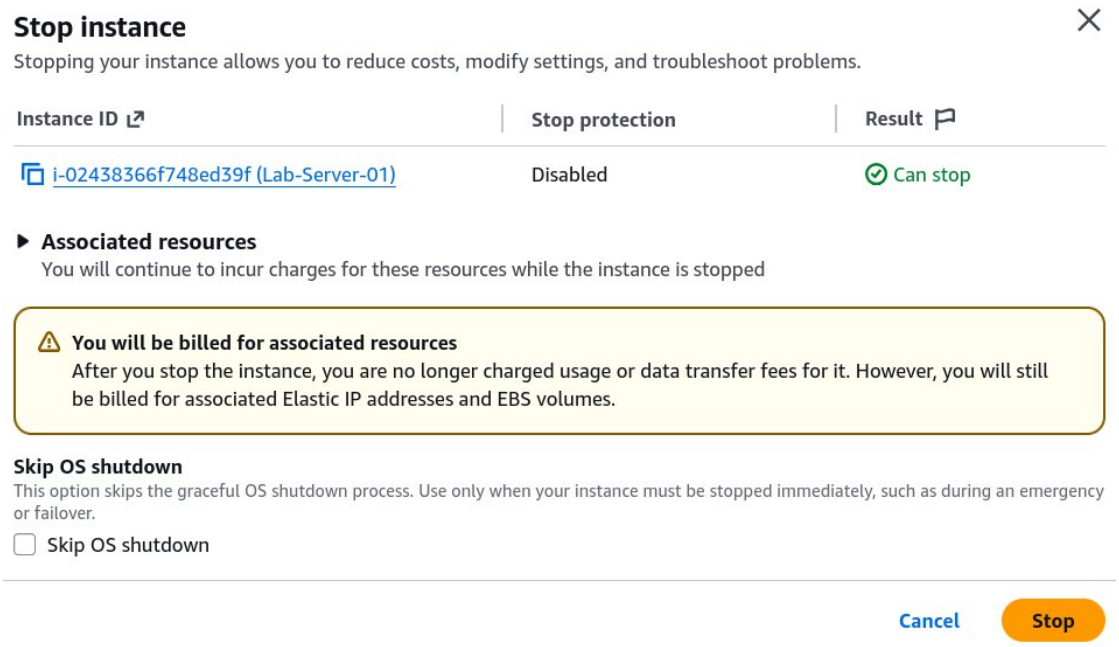

Step 3: Power Operations (The Lifecycle)

Perform the following actions and watch the "Instance State" column:

1. Reboot: Equivalent to restarting your computer. The Public IP stays the same.
2. Stop: The "Power Off" state. You stop paying for Compute, but still pay for EBS Storage.
    - Warning: When you Start a stopped instance, it usually receives a new Public IP address.
3. Terminate: Deletes the instance permanently. This cannot be undone.

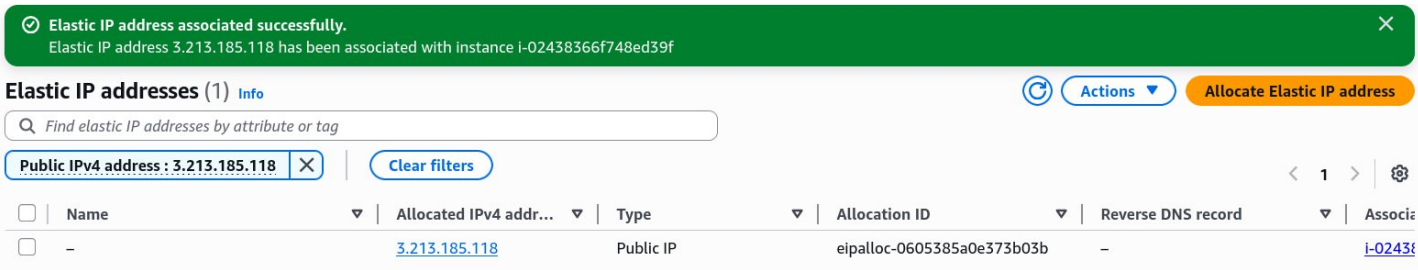


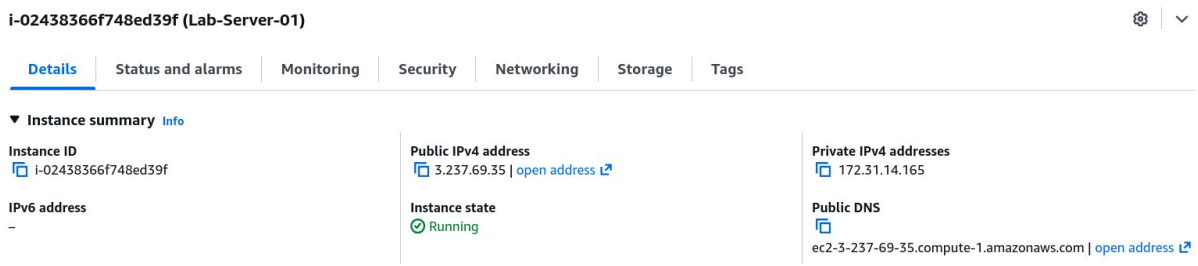


Step 4: Explore Elastic IPs

Since Public IPs change when an instance is stopped/started, we use Elastic IPs for a persistent address.

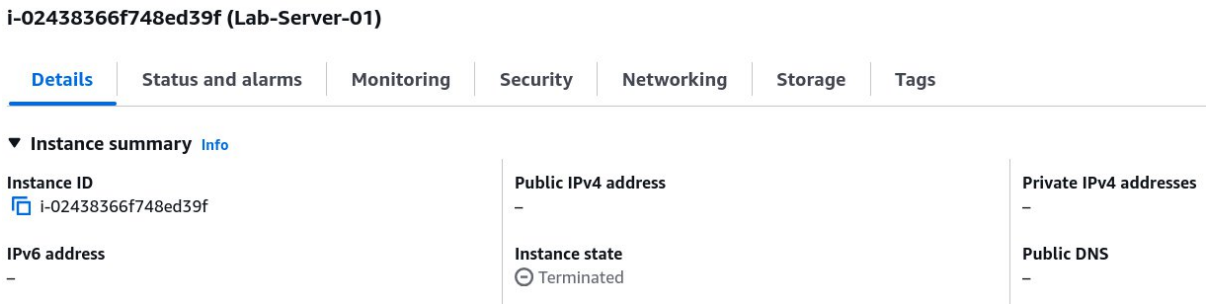
1. In the left sidebar, go to Network & Security > Elastic IPs.
2. Click Allocate Elastic IP address > Allocate.
3. Select the new IP > Actions > Associate Elastic IP address.
4. Choose your Lab-Server-01 and click Associate.
5. Verify: Stop and Start your instance. Notice the Public IP now remains exactly the same.





Public IP vs. Elastic IP

Feature	Default Public IP	Elastic IP
Persistence	Lost when an instance is stopped.	Remains with you until manually released.
Cost	Free while the instance is running.	Free if associated with a running instance.
Idle Fee	N/A	Charged a small fee if allocated but not attached to a running instance.

Cleanup (Critical Step)

1. Terminate Instance: Select instance > Instance state > Terminate.
2. Release Elastic IP: You must manually Disassociate and then Release the Elastic IP, or AWS will continue to charge you for "occupying" that IP address while it's idle.